

Ex2 - Problème du sac à dos bi-objectifs

Q2.1

$$x_0 = (1, 1, 1, 1, 1, 0, 0, 0)$$

$$U(x_0) = (28, 37)$$

$$\sum w_i x_i = 18$$

$$W = 20$$

$$w = (4, 3, 2, 1, 8, 7, 6, 5)$$

$$p_1 = (11, 4, 5, 1, 7, 13, 4, 3)$$

$$p_2 = (9, 8, 2, 2, 16, 5, 9, 4)$$

$$N(x_0) = \left\{ (0, 1, 1, 1, 1, 0, 1, 0), \quad \sum = 20, \quad y = (24, 37) \right. \\ \left. \text{pas ajoutée} \right\}$$

$$(0, 1, 1, 1, 1, 0, 0, 1), \quad \sum = 19, \quad y = (20, 32)$$

$$(1, 0, 1, 1, 1, 0, 0, 1), \quad \sum = 20, \quad y = (27, 33)$$

$$(1, 1, 1, 1, 0, 1, 0, 0), \quad \sum = 17, \quad y = (34, 26)$$

$$(1, 1, 1, 1, 0, 0, 1, 0), \quad \sum = 16, \quad y = (25, 30)$$

$$(1, 1, 1, 1, 0, 0, 0, 1), \quad \sum = 15, \quad y = (24, 25) \quad \left\{ \right.$$

$$P = \{(34, 26)\} \quad Y_E^* = \{(28, 37), (34, 26)\}$$

$$x_c = (1, 1, 1, 1, 0, 1, 0, 0) \quad \sum x_i w_i = 17 \quad y = (34, 26)$$

$$N(x_c) = \left\{ (0, 1, 1, 1, 0, 1, 1, 0), \quad \sum = 19 \quad y = (27, 26) \right.$$

$$(0, 1, 1, 1, 0, 1, 0, 1), \quad \sum = 18 \quad y = (26, 24)$$

$$(1, 0, 1, 1, 0, 1, 1, 0), \quad \sum = 20 \quad y = (34, 27)$$

$$(1, 0, 1, 1, 0, 1, 0, 1), \quad \sum = 19 \quad y = (33, 22)$$

$$(1, 1, 0, 1, 0, 1, 0, 1) \quad \sum = 20 \quad y = (32, 28)$$

$$(1, 1, 1, 1, 1, 0, 0, 0) \quad \sum = 18 \quad y = (28, 37) \quad \text{déjà dans } Y_E$$

$$(1, 1, 1, 1, 0, 0, 1, 0) \quad \sum = 16 \quad y = (25, 30)$$

$$(1, 1, 1, 1, 0, 0, 0, 1) \quad \sum = 15 \quad y = (24, 25) \quad \left\{ \right.$$

$$P = \{(34, 27), (32, 28)\} \quad Y_E^* = \{(28, 37), (34, 27), (32, 28)\}$$

$$x_c = (1, 0, 1, 1, 0, 1, 1, 0), \quad \xi = 20, \quad y = (34, 27)$$

$$N(x_c) = \left\{ \begin{array}{ll} (0, 1, 1, 1, 0, 1, 1, 0), & \xi = 19, \quad y = (27, 26) \\ (1, 1, 1, 1, 0, 0, 1, 0) & \xi = 16, \quad y = (25, 30) \\ (1, 0, 1, 1, 0, 0, 1, 1) & \xi = 18, \quad y = (24, 26) \\ (1, 1, 1, 1, 0, 1, 0, 0) & \xi = 17, \quad y = (34, 26) \\ (1, 0, 1, 1, 0, 1, 0, 1) & \xi = 19, \quad y = (33, 22) \end{array} \right\}$$

$$P = \{(32, 28)\} \quad \tilde{Y}_E = \{(28, 37), (34, 27), (32, 28)\}$$

$$x_c = (1, 1, 0, 1, 0, 1, 0, 1), \quad \xi = 20, \quad y = (32, 28)$$

$$N(x_c) = \left\{ \begin{array}{ll} (0, 1, 1, 1, 0, 1, 0, 1), & \xi = 18, \quad y = (26, 21) \\ (1, 0, 1, 1, 0, 1, 0, 1), & \xi = 19, \quad y = (33, 22) \text{ per } (34, 27) \\ (1, 1, 0, 1, 0, 0, 1, 1), & \xi = 19, \quad y = (23, 32) \text{ per } (28, 37) \\ (1, 1, 1, 1, 0, 0, 0, 1), & \xi = 15, \quad y = (24, 25) \\ (1, 1, 1, 1, 0, 1, 0, 0), & \xi = 17, \quad y = (34, 26) \end{array} \right\}$$

$$P = \{ \}$$

$$\Rightarrow \tilde{Y}_E = \{(28, 37), (34, 27), (32, 28)\}$$

Q2.2

$$x^2 = (1, 0, 0, 1, 1, 1, 0, 0) \quad \{x_i w_i = 20 \quad y = (32, 32)$$

$$N(x^2) = \left\{ \begin{array}{lll} (0, 1, 0, 1, 1, 1, 0, 0) & \xi = 19 & y = (\cancel{25}, 31), \\ (0, 0, 1, 1, 1, 1, 0, 0) & \xi = 18 & y = (\cancel{26}, \cancel{25}) \\ (1, 1, 1, 1, 0, 1, 0, 0) & \xi = 17 & y = (\cancel{34}, \cancel{26}) \\ (1, \cancel{1}, 1, 1, 0, 1, 0, 0) & \xi = 17 & y = (\cancel{34}, \cancel{26}) \\ (1, 0, 1, 1, 0, 1, 1, 0) & \xi = 20 & y = (34, 27) \\ (1, 1, 0, 1, 0, 1, 0, 1) & \xi = 20 & y = (\cancel{32}, \cancel{28}) \\ (1, 1, 1, 1, 1, 0, 0, 0) & \xi = 18 & y = (28, 37) \\ (1, 1, 1, 1, 1, 0, 0, 0) & \xi = 18 & y = (\cancel{28}, \cancel{37}) \\ (1, 0, 0, 1, 1, 0, 1, 0) & \xi = 19 & y = (\cancel{23}, \cancel{36}) \\ (1, 0, \cancel{1}, 1, 1, 0, 0, 1) & \xi = 20 & y = (\cancel{27}, \cancel{33}) \end{array} \right.$$

$$P = \{(34, 27), (28, 37)\} \quad Y_E^2 = \{(32, 32), (34, 27), (28, 37)\}$$

$$x_c = (1, 0, 1, 1, 0, 1, 1, 0) \rightarrow \text{ne donne rien (voir Q2.1)}$$

$$x_c = (1, 1, 1, 1, 1, 0, 0, 0) \rightarrow \text{ne va rien donner (voir Q2.1)}$$

$$\Rightarrow Y_E^2 = \{(32, 32), (34, 27), (28, 37)\}$$

Q2.3

Comme $\tilde{Y}_E^2 \triangleleft \tilde{Y}_E^1$ (chaque solution de $\tilde{Y}_E^1$ est faiblement dominée par au moins une solution de $\tilde{Y}_E^2$ et $\tilde{Y}_E^2 \neq \tilde{Y}_E^1$), $\tilde{Y}_E^2$ est la meilleure des deux approximations.